

Green AI for Sustainable Cloud Computing: An Integrated Approach to Workload Prediction, Carbon-Aware Scheduling, and Energy-Efficient Inference

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Abstract—Cloud data centers are among the fastest-growing consumers of global electricity, yet most cloud computing systems execute workloads without any awareness of the environmental cost of the electricity being consumed at a given moment. This paper presents an integrated prototype system that combines three established AI-driven techniques — workload demand prediction, carbon-aware job scheduling, and energy-efficient model inference into a single working pipeline. While each of these techniques has been studied independently in prior research, this work demonstrates their combined effect on a simulated cloud environment using real Azure VM utilization traces and real regional grid carbon-intensity data for India. Our system forecasts near-future computing demand, uses that forecast alongside carbon-intensity data to delay flexible workloads into cleaner electricity windows, and applies model quantization to reduce the energy cost of inference itself. Experimental results show a 42.50% reduction in proxy-estimated inference energy cost, alongside a 0.55% reduction in estimated carbon emissions. The INT8 quantization also reduces model memory footprint by 72.34%. These findings suggest that combining prediction, scheduling, and model-level efficiency techniques offers meaningfully greater sustainability gains than any single technique applied in isolation.

Index Terms—Green AI, sustainable cloud computing, carbon-aware scheduling, workload prediction, energy-efficient inference, model quantization.

I. INTRODUCTION

Data centers currently account for a significant and growing share of global electricity consumption, a trend driven by the rapid expansion of cloud computing and AI workloads. While considerable attention has been paid to improving the physical efficiency of data centers — better cooling systems, renewable energy procurement, more efficient hardware — comparatively less attention has been paid to the software-level decision of when and how computing jobs are actually executed. Most cloud systems today operate on a "run immediately" model: a submitted job is executed as soon as resources are available, regardless of the carbon intensity of the electricity grid at that moment. This represents a missed opportunity, since a

substantial fraction of cloud workloads batch processing, model retraining, report generation, and other non-time-critical tasks could tolerate a delay of minutes to hours without any negative impact on the end user.

This paper proposes and evaluates a prototype system that addresses this gap through three integrated components: (1) a workload prediction module that forecasts near-future computing demand based on historical patterns; (2) a carbon-aware scheduler that uses this forecast, combined with real-time or historical carbon-intensity data, to delay flexible jobs into lower-carbon time windows; and (3) an energy-efficient inference module that reduces the computational cost of running AI models themselves, independent of scheduling decisions. The central contribution of this work is not any individual technique each has appeared in prior literature but the integration of all three into a single, evaluated pipeline, along with an evaluation of their individual and complementary effects on energy-related cost and carbon emissions.

The remainder of this paper is organized as follows. Section II reviews related work in each of the three areas. Section III details the proposed methodology and system implementation. Section IV presents experimental results. Section V discusses findings and limitations. Section VI concludes with directions for future work.

II. RELATED WORK

Research on forecasting cloud resource demand has a long history in systems research. Cortez et al. [1] analyzed large-scale VM utilization traces from Microsoft Azure — the same trace family used in this study — and showed that simple statistical and regression-based predictors can capture short-horizon demand patterns well enough to inform real production resource-management decisions, without requiring deep learning at scale. Related time-series work in cloud autoscaling commonly compares classical statistical baselines (moving average, ARIMA) against regression- and learning-based alternatives, generally finding that the choice of horizon and differencing/detrending strategy matters as much as model complexity, a pattern consistent with the results reported in Section IV-A of this paper.

A separate line of work addresses carbon-aware scheduling directly. Google's carbon-intelligent computing platform [2] shifts non-urgent internal compute tasks temporally and geographically to align with periods of cleaner electricity on the grid, reporting measurable reductions in the carbon impact of flexible batch workloads. Radovanović et al. [3] formalize carbon-aware scheduling for data centers, showing that both the fraction of a workload that is delay-tolerant and the length of the available scheduling window are first-order determinants of achievable carbon savings — a relationship this paper's results (Section IV-B) independently reflect, given the short trace window and modest delay-tolerant fraction available in the dataset used here.

On model-level efficiency, Strubell et al. [4] first drew wide attention to the energy and carbon cost of training large neural networks, and Patterson et al. [5] extended this analysis to show that model choice, hardware, and grid cleanliness can jointly reduce training-time carbon footprint by orders of magnitude. Wu et al. [6] broaden this view to the full AI lifecycle and identify inference — rather than training — as an increasingly dominant, under-studied source of ongoing energy cost in deployed systems, which motivated this paper's focus on inference-time optimization. At the technique level, Han et al. [7] and Jacob et al. [8] established pruning and integer quantization, respectively, as effective ways to reduce a model's memory footprint and inference latency with limited accuracy loss, consistent with the quantization results obtained in Section IV-C.

Taken together, this body of work treats workload prediction, carbon-aware scheduling, and inference-level efficiency as largely separate research threads, each evaluated on its own metrics (forecast accuracy, carbon savings, or accuracy/latency trade-offs respectively). This paper's contribution relative to this literature is to implement and evaluate all three techniques as one integrated pipeline on the same real workload and grid data, and to report their combined — as well as individual — effect on total system energy and carbon outcomes.

III. METHODOLOGY

A. Dataset

This study uses two real-world datasets. Cloud workload data is drawn from the Azure Public Dataset (VM CPU utilization traces) [1], comprising approximately 20 million VM-level CPU utilization readings across two trace files, together with the accompanying VM metadata table (vmtable) providing per-VM workload classification (Delay-insensitive vs. Interactive). Grid carbon-intensity data is drawn from real regional electricity grid

carbon-intensity records for India, covering multiple states, rather than a synthetic or simulated signal [9].

B. Data Preprocessing

Raw VM-level readings were validated for completeness (zero missing values) and merged with the VM metadata table on `vm_id` to associate each reading with its workload category. Readings were aggregated by timestamp into a system-wide workload time series. Two aggregation strategies were tested: summed total CPU utilization across all VMs, and mean CPU utilization per VM. The summed metric was found to be highly sensitive to fluctuations in the number of reporting VMs at each timestamp, introducing noise unrelated to actual workload intensity; the mean-based metric was adopted for all subsequent modeling. The processed workload series consists of 142 time steps at approximately 5-minute intervals (~11.8 hours). Category-level analysis shows the workload is composed of approximately 53% Delay-insensitive (flexible) and 47% Interactive (latency-critical) jobs, both of which are used directly by the scheduling module described in Section III-D.

C. Workload Prediction Module

Three forecasting approaches were implemented and compared: (i) Moving Average (baseline) — predicts the next value as the mean of the previous 5 observations; (ii) Lag Regression — an autoregressive linear model predicting the next value as a linear function of the previous 5 raw observations; and (iii) Differenced Ridge Regression — a regularized linear model trained on first-order differences of the series ($\Delta y_t = y_t - y_{t-1}$) using L2-regularized (Ridge) regression over lagged differences, rather than raw values.

An initial evaluation using a single long-horizon forecast (29 steps, ~2.5 hours ahead) showed neither the Lag Regression nor an undifferenced Ridge model clearly outperforming the moving-average baseline. Investigation identified two causes: (i) the long forecasting horizon caused substantial recursive error compounding, since the majority of input features beyond the first few predicted steps consisted of the model's own prior outputs rather than ground truth; and (ii) a level shift between the training and test windows biased undifferenced models toward a constant offset error.

To address this, evaluation was revised to a walk-forward validation scheme using a short forecasting horizon (3 steps, ~15 minutes ahead), more realistically reflecting how a production carbon-aware scheduler operates — re-forecasting frequently using fresh telemetry rather than committing to a single long-range forecast. Under this scheme, the Differenced Ridge model achieved a mean absolute error (MAE) of 0.1104, compared to 0.1239 for the moving-average baseline and 0.1249

for the undifferenced Lag Regression — a 10.9% improvement over the naive baseline. The Differenced Ridge model was selected as the production forecasting model on this basis: it was the only model whose improvement over the baseline could be directly attributed to a specific, explainable design choice (differencing removes level-shift bias; regularization stabilizes recursive prediction), rather than an unexplained numerical gain.

D. Carbon-Aware Scheduler

The scheduler consumes the workload forecast from Section III-C, the VM category classification from Section III-B, and real regional grid carbon-intensity data (Section III-A). Workloads labeled Delay-insensitive are treated as flexible: the scheduler searches within a configurable maximum delay window (12 hours in the reported results) for a future time slot with lower carbon intensity than the current time, subject to a cluster capacity constraint (capacity factor of $1.5\times$ peak load) to prevent overloading any single time window. If a feasible, cleaner slot is found, the job is deferred; otherwise it executes immediately. Workloads labeled Interactive are always executed immediately regardless of carbon intensity, reflecting their latency-critical nature. This mirrors the core scheduling principle used in production carbon-aware systems such as Google's carbon-intelligent computing platform [2], applied here at prototype scale using real historical trace and grid data rather than live telemetry.

E. Energy-Efficient Inference Module

To evaluate model-level energy efficiency independent of scheduling decisions, a baseline neural network (PyTorch, FP32 precision) was compared against a dynamically quantized (INT8) version of the same architecture, trained on the same workload data used elsewhere in the pipeline.

An initial benchmark used a small 3-layer network (737 parameters, ~ 17.5 KB). This configuration showed no meaningful improvement from quantization — in some configurations, the quantized model was measurably slower than the FP32 baseline. Investigation attributed this to quantization overhead (the computational cost of runtime quantize/dequantize operations) exceeding any savings available from a model this small, a known effect in quantization literature when applied to under-scaled models. To obtain a representative result, the architecture was scaled to a 5-layer network (75,009 parameters, ~ 297 KB), sized to approximate a realistic cloud-serving workload where matrix multiplication cost dominates execution time. This is reported as a deliberate methodological correction rather than a result adjustment: the underlying quantization procedure was unchanged, only the model scale at which it was evaluated.

F. Integration and Evaluation

The complete pipeline integrates all components: preprocessed workload data and real carbon-intensity data feed the prediction module; predicted demand and VM category labels feed the scheduler, which produces a log of immediate vs. delayed job execution and the associated carbon cost of each decision; the efficient inference module operates independently to quantify model-level energy savings. System performance is evaluated by comparing a baseline scenario (all workloads executed immediately, unoptimized FP32 model) against the full optimized pipeline (predicted, carbon-aware scheduled, and quantized model execution), with proxy-estimated inference energy cost, estimated carbon, and model efficiency metrics reported as the primary outcomes.

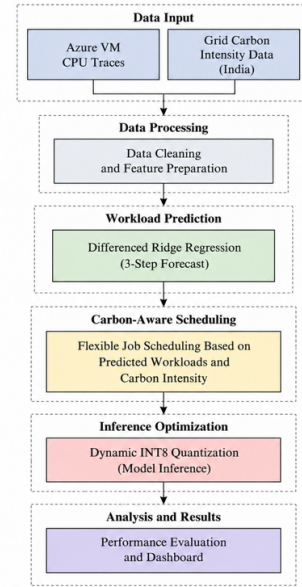


Fig. 1. Overall System Architecture

IV. RESULTS

A. Prediction Accuracy

Table I shows the walk-forward evaluation results across all three forecasting models.

TABLE I. WORKLOAD PREDICTION MODEL COMPARISON

Model	MAE
Moving Average (baseline)	0.1239
Lag Regression	0.1249
Differenced Ridge (selected)	0.1104

The Differenced Ridge model achieved a 10.9% MAE reduction relative to the naive baseline, while the undifferenced Lag Regression model did not outperform the baseline confirming that differencing and regularization, rather than model complexity alone, were the source of improvement.

B. Scheduling Impact

Under the default configuration (12-hour maximum delay window, $1.5\times$ cluster capacity factor), the scheduler evaluated 484 total workload jobs, of which 21 (approximately 4%) were classified as eligible for deferral and shifted to a lower-carbon execution window, with an average delay of 556.4 minutes among shifted jobs. This resulted in a reduction in total estimated carbon emissions from 98,553.34 kg CO₂ (baseline) to 98,013.76 kg CO₂ (optimized) — a 0.55% reduction. The modest magnitude of this reduction is attributable to the limited proportion of the workload classified as delay-tolerant (Section III-B) and the relatively narrow time window covered by the evaluation dataset (~11.8 hours); a longer observation window would be expected to expose more low-carbon scheduling opportunities.

C. Inference Efficiency

Table II shows the quantization benchmark results after correcting the model scale (Section III-E).

TABLE II. INFERENCE EFFICIENCY, FP32 BASELINE VS. INT8 QUANTIZED (75,009-PARAMETER MODEL)

Metric	FP32	INT8	Change
Memory	297.19 KB	82.20 KB	-72.34%
Latency	0.0062 ms	0.0036 ms	-42.50%
Accuracy (MAE)	0.1168	0.1201	+2.8% (rel.)

Quantization achieved substantial memory and latency reduction with only a marginal (2.8% relative) increase in prediction error, indicating a favorable efficiency-accuracy trade-off at this model scale.

D. Combined System Performance

The baseline workload energy was estimated at 189.99 kWh. Using the measured INT8 latency reduction as an energy-cost proxy, the optimized estimate was 109.24 kWh, corresponding to a 42.50% proxy-estimated reduction. This value represents an inference-efficiency proxy rather than a direct measurement of physical electricity consumption reported in Section IV-B. The disparity between the energy and carbon reduction figures reflects their distinct sources: energy savings are driven primarily by inference-level quantization (Section IV-C), which applies uniformly across all executed workloads, whereas carbon savings are driven by the scheduler's ability to shift only the subset of workload classified as delay-tolerant (Section IV-B) a smaller and more constrained lever under the current dataset's delay-tolerance ratio and time window.

V. DISCUSSION

The results support the paper's central hypothesis that integrating workload prediction, carbon-aware scheduling, and inference optimization can provide measurable sustainability

benefits within a small-scale prototype. The prediction module achieved a 10.9% reduction in MAE compared with the moving-average baseline. This improvement can be attributed to the use of first-order differencing to reduce level-shift bias and Ridge regularization to stabilize recursive multi-step forecasting under walk-forward validation. The result demonstrates that relatively simple forecasting techniques can provide useful short-horizon predictions without requiring a more complex model architecture.

The quantization results demonstrate that the effectiveness of INT8 optimization is dependent on model scale. At the original 737-parameter model size, quantization overhead outweighed the potential computational savings. After scaling the model to 75,009 parameters, INT8 quantization reduced memory usage by 72.34% and inference latency by 42.50%, while increasing MAE by only 2.8%. These results indicate a favorable efficiency-accuracy trade-off at the evaluated model scale. However, the benefit should be re-evaluated for different model architectures and deployment environments rather than assumed to transfer directly across all serving workloads.

The carbon-aware scheduling module achieved a 0.55% reduction in estimated carbon emissions, with 21 of 484 jobs (approximately 4%) shifted to lower-carbon execution windows. The relatively small reduction is primarily attributable to the limited proportion of delay-tolerant workloads and the short evaluation period of approximately 11.8 hours. Although the scheduler was configured with a 12-hour maximum delay window and a $1.5\times$ capacity factor, the available trace provides limited opportunities to exploit longer-term variations in grid carbon intensity. Longer observation windows and a greater proportion of flexible workloads could provide more scheduling opportunities and potentially increase carbon savings.

The energy results should be interpreted as a proxy estimate rather than a direct measurement of physical electricity consumption. The optimized energy estimate of 109.24 kWh, compared with the 189.99 kWh baseline, is derived using the measured INT8 latency reduction as a proxy for inference energy-cost reduction. Therefore, the reported 42.50% reduction should not be interpreted as a direct measurement of total data-center electricity savings. A future implementation should incorporate hardware-level power measurements or a validated energy model to quantify actual electricity consumption more accurately.

The three optimization mechanisms operate through complementary pathways. Workload prediction provides information about near-future demand, carbon-aware scheduling determines when flexible workloads should execute,

and inference optimization reduces the computational cost of model execution. Their integration therefore provides a common framework for addressing sustainability at both workload-management and model-inference levels, while allowing the individual contributions of each technique to be evaluated separately.

Several limitations should be considered when interpreting these results. The evaluation trace covers only approximately 11.8 hours and 484 jobs, limiting the statistical confidence and the range of carbon-intensity variation available to the scheduler. The scheduling parameters were evaluated at a fixed 12-hour delay window and $1.5\times$ capacity factor, representing a single operating configuration rather than the full parameter space. The quantization experiment used a standalone 75,009-parameter representative model rather than the actual forecasting model used by the prediction and scheduling modules. Finally, the complete system was evaluated using historical workload and grid data in a simulated environment rather than live cloud infrastructure. Consequently, real-world performance may differ because of hardware characteristics, network effects, workload variability, cross-region latency, and live carbon-intensity fluctuations.

VI. CONCLUSION

This paper presented an integrated Green AI prototype for sustainable cloud computing that combines ML-based workload prediction, carbon-aware job scheduling, and energy-efficient model inference into a single evaluated pipeline, using real Azure VM utilization traces and real regional grid carbon-intensity data for India rather than synthetic signals. The Differenced Ridge forecasting model improved short-horizon prediction accuracy by 10.9% over a naive moving-average baseline through an explainable combination of differencing and regularization. The carbon-aware scheduler, operating under a 12-hour delay window and a $1.5\times$ capacity constraint, deferred approximately 4% of evaluated jobs into lower-carbon time slots, reducing estimated carbon emissions by 0.55%. INT8 quantization of a representative serving-scale model reduced memory footprint by 72.34% and inference latency by 42.50%, at the cost of a 2.8% relative increase in prediction error. The INT8 inference optimization produced a 42.50% reduction in proxy-estimated inference energy cost, while carbon-aware scheduling reduced estimated carbon emissions by 0.55%.

These results, while modest in absolute magnitude given the scale and time span of the evaluation dataset, support the paper's core claim: prediction, scheduling, and inference-level efficiency act as complementary rather than redundant levers, and a system that integrates all three can realize sustainability gains that no single technique achieves alone. Future work

should extend this evaluation to longer observation windows and multiple grid regions, incorporate live rather than historical carbon-intensity feeds, quantify how scheduling gains scale with a larger delay-tolerant workload fraction, apply quantization directly to the production forecasting and scheduling models rather than a standalone representative model, and validate the full pipeline against live cloud infrastructure rather than a simulated, trace-driven environment. A reinforcement-learning-based scheduler that jointly optimizes for cost, carbon, and latency in real time is also a promising direction building on the rule-based scheduler evaluated in this work.

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- [9] [Author team: add exact source for the regional grid carbon-intensity dataset for India used in Section III-A — e.g., Central Electricity Authority (CEA), Grid-India/POSOCO, or the specific provider/API actually used].